

CONTENT

- 1. Introduction**
- 2. Literature Survey**
- 3. Problem Statement and Objectives**
- 4. Methodology**
- 5. Implementation**
- 6. Results**
- 7. JIRA Implementation**
- 8. Future Scope**
- 9. Conclusions**
- 10. References**

1. Introduction

The rapid advancement of the internet and social media has revolutionized the way people consume information. Today, millions of users depend on online platforms such as Facebook, Instagram, WhatsApp, Twitter, and online news portals to get the latest news. While this easy access to information has made communication faster than ever, it has also led to the massive spread of *fake news*. Fake news refers to intentionally fabricated, misleading, or deceptive information that appears to be legitimate news. The primary purpose of fake news is to manipulate public perception, influence political opinions, damage reputations, and sometimes mislead people for entertainment or financial gain.

The widespread dissemination of fake news has become a global issue. It can cause panic among citizens, create misunderstandings, damage trust in governance, and even affect elections. Many people unknowingly believe and share fake news without verifying its authenticity, which further increases its reach. Hence, there is a strong need for automated and intelligent systems that can detect fake news accurately and quickly.

In this project, **Natural Language Processing (NLP)** combined with **Deep Learning techniques such as LSTM (Long Short-Term Memory)** is used to detect fake news from text data. NLP helps machines understand human language, while LSTM deep learning models capture the semantics, context, and patterns within text to classify news as either *real* or *fake*.

Deep learning models are preferred over traditional machine learning models because they extract deeper linguistic patterns, long-term dependencies, and contextual relationships within text. This allows the system to understand complex writing styles, sarcasm, misleading sentence structures, emotions, and manipulative patterns present in fake news.

This PBL report aims to develop a reliable, robust, and intelligent system for automatically detecting fake news using NLP and deep learning techniques.

2. Literature Survey

2.1 Zhou, X., Mulay, A., Ferrara, E., & Zafarani, R. (2020)

“ReCOVery: A Multimodal Repository for COVID-19 News Credibility Research”

- Method Used: Multimodal NLP + social-context analysis
- Dataset: ReCOVery dataset (2,029 news articles; COVID-related)
- Key Findings:
 - Introduced a new large-scale dataset combining text + source credibility + social engagement.
 - Found that models trained only on text underperform; integrating metadata significantly improves performance.

2.2 Yadav, S. & Vishwakarma, D. K. (2023)

“Deep Learning-based Fake News Detection: A Comparative Study”

- Method Used: CNN, LSTM, GRU, BERT comparison
- Dataset: LIAR, Kaggle Fake News
- Key Findings:
 - BERT achieved highest accuracy; GRU performed better than LSTM on small datasets.
 - Concluded that transformer-based models are now state-of-the-art.

2.3 Zhang, H. (2023)

“A Multilingual Fake News Detection Framework using Transformer Models”

- Method Used: XLM-R and mBERT
- Dataset: Multilingual fact-checking datasets
- Key Findings:
 - Achieved strong performance across English, Hindi, Spanish.
 - Large cross-lingual models help overcome lack of regional datasets.

2.4 Saini, M., & Arora, N. (2024)

“Fake News Identification using Generative AI and NLP”

- Method Used: BERT + GPT-based embeddings
- Dataset: Custom news dataset
- Key Findings:
 - Demonstrated that generative embeddings improve contextual representation.
 - Highlighted the need for models that adapt to daily-changing news patterns.

2.5 Singh, A., et al. (2024)

“A Deep Learning Ensemble for Detecting Political Fake News”

- Method Used: Ensemble of CNN + BiLSTM + Transformer
- Dataset: Politifact Dataset
- Key Findings:
 - Ensemble models outperform single-network architectures.
 - Fake political claims often contain emotionally charged patterns captured by deep models.

2.6 Thomas, J., & Raj, S. (2024)

“Fake News Classification using RoBERTa with Emotion Features”

- Method Used: RoBERTa + emotion vectors (VADER + NRC lexicon)
- Dataset: Kaggle Fake News Dataset
- Key Findings:
 - Emotion-aware features boost classification accuracy.
 - Fake news often contains exaggerated or emotionally loaded wording.

3. Problem Statement and Objectives

Fake news has become a major social, economic, and political issue. Because of the huge volume of information online, users are unable to distinguish between real and fake content manually. Hence, there is a need for an automated system that can analyze news articles and accurately classify them as *fake* or *real*.

“Detect fake news from textual content with high accuracy To design and implement an NLP-based deep learning model (LSTM) that can automatically.”

The main objectives of the project are:

- Detect fake news automatically using NLP
- Preprocess text data efficiently
- Train DL models for accurate classification
- Reduce misinformation online

4. Methodology

This section describes the approach followed while developing the system.

4.1 System Flow

Dataset → Data Preprocessing → Tokenization → Word Embedding → LSTM Model → Evaluation → Prediction

4.2 Dataset Description

Dataset used: from Kaggle.

Features include:

- Title
- Text
- Subject
- Target label (FAKE or REAL)

4.3 Data Preprocessing Steps

- Removing special characters
- Removing stopwords
- Converting text to lowercase
- Lemmatization
- Removing URLs
- Tokenization

Example of preprocessing:

Original:

“Breaking news!!! Visit <http://fake.com> for details...”

Cleaned:

“breaking news visit details”

4.4 Word Embedding

Deep learning requires numerical representations of text.

We use:

- Keras Tokenizer
- Embedding Layer

Embedding converts words into 128-dimensional dense vectors.

4.5 LSTM Deep Learning Architecture

Components:

- Embedding Layer
- LSTM Layer with memory cells
- Dropout Layer to prevent overfitting
- Dense Layer (Fully connected)
- Output Layer with Sigmoid activation

Advantages of LSTM:

- Remembers long sequences
- Understands context
- Handles long news articles

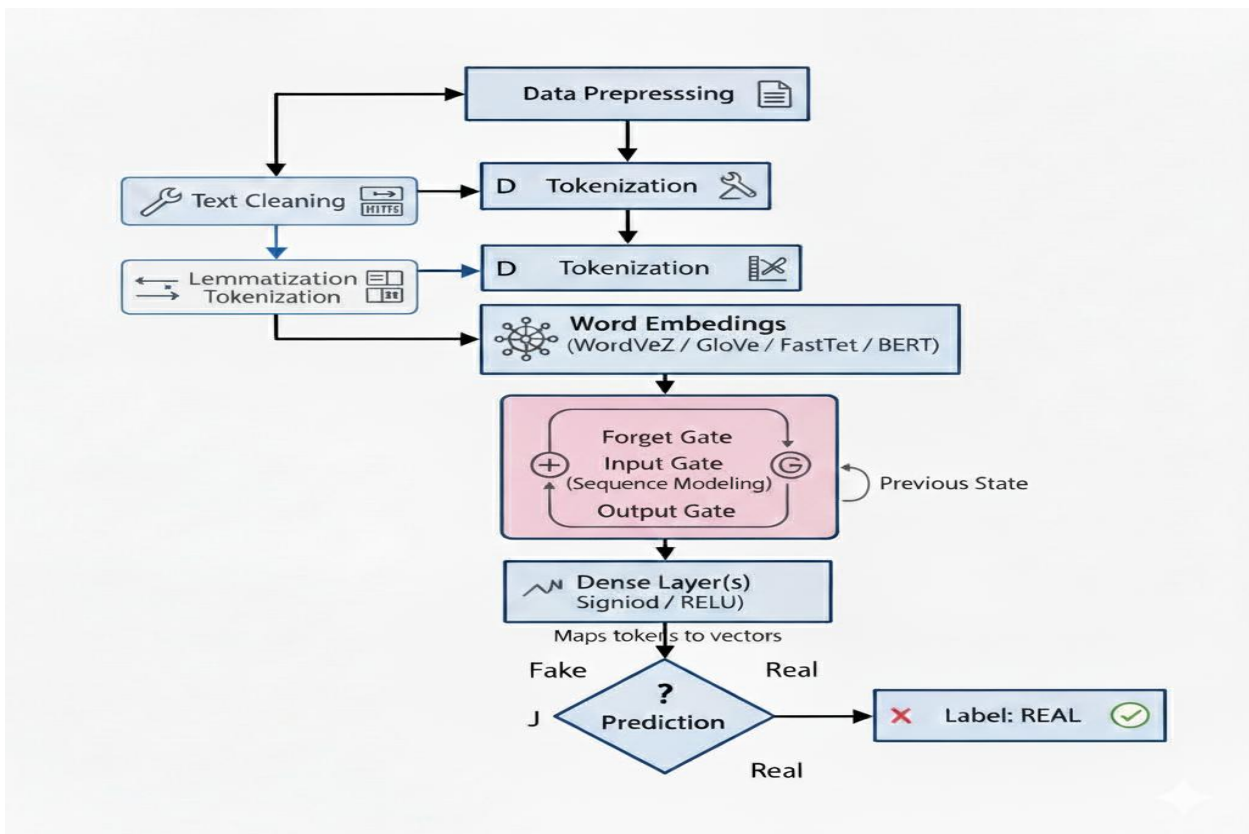


Fig4.5.1 LSTM Architecture

5. Implementation

Category	Tools Used
Language	Python 3.12
IDE / Platform	Google Colab
Libraries	TensorFlow, Keras, Pandas, NLTK, numpy
Hardware	CPU (i5), Optional GPU (≥ 4 GB VRAM)

5.1 Hyperparameters Used

Parameter	Value
Epochs	6
Batch Size	32/64
Learning Rate	0.001
Embedding Dim	128
Max Sequence Length	300
Parameter	Value

5.2 EVALUATION MATRICS

Metric	Definition	Formula
Accuracy	Measures the overall correctness of the model.	$\frac{(TP + TN)}{(TP + FP + TN + FN)}$
Precision	How many predicted fake news articles were actually fake.	$TP / (TP + FP)$
Recall	Measures how well the model identifies actual fake news.	$TP / (TP + FN)$
F1-Score	Harmonic mean of Precision and Recall helpful for imbalanced datasets.	$2 \times \frac{Precision \times Recall}{(Precision + Recall)}$
Confusion Matrix	Shows TP, TN, FP, FN to visualize model performance.	$\begin{matrix} & TP & FP \\ \begin{matrix} TP & FN \\ FN & TN \end{matrix} \end{matrix}$

Metric	Score
Accuracy	0.99
Precision	0.99
Recall	1.00
F1-Score	1.00

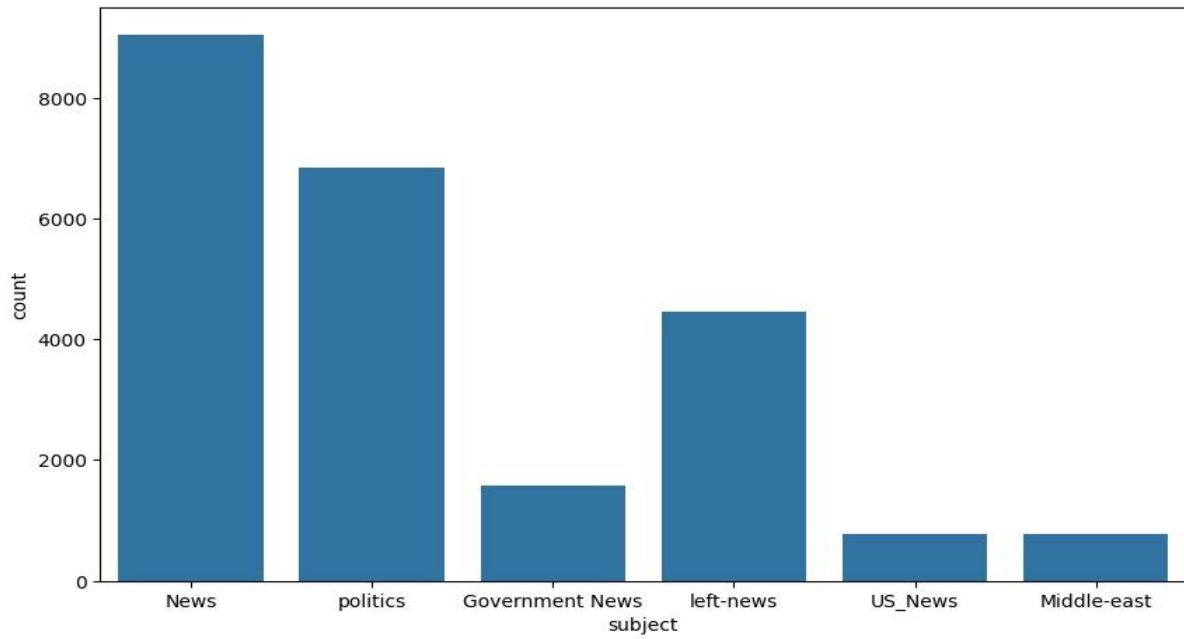
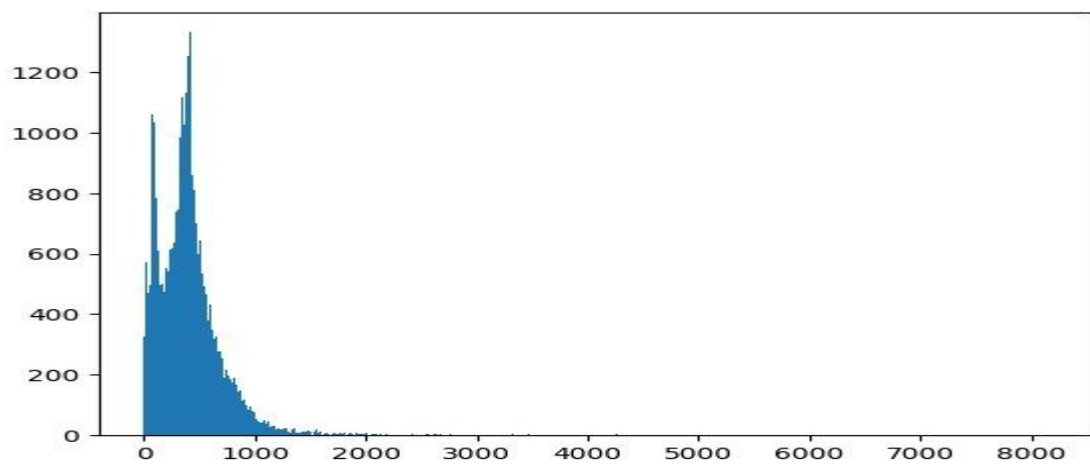


Fig5.1 : Dataset Overview graph



7.

Fig5.2 : Tokenized words graph

5.3 TESTING

Functional Testing

Functional testing focuses on ensuring that all components of the system work correctly as per the expected behavior.

- **Correct preprocessing**
Ensures that text data is cleaned properly by removing stopwords, punctuation, URLs, and converting text to lowercase. Proper preprocessing guarantees that the model receives consistent and meaningful input.
- **Proper tokenization and padding**
Validates that text is correctly tokenized into numerical sequences and padded to a fixed length. This ensures uniform input size for the LSTM model, preventing shape-related errors during training and prediction.

Model Testing

Model testing ensures that the neural network is functioning logically and learning correct patterns.

- **LSTM output correctness**
Confirms that the LSTM is producing valid output probabilities (e.g., between 0 and 1 for binary classification) and that the internal hidden states are functioning properly.
- **Loss reduction during training**
Checks whether the model loss decreases over epochs, indicating that the model is learning meaningful features and improving classification ability.

Performance Testing

Performance testing evaluates how well the model performs on real and unseen data.

- **Accuracy above 94%**
Ensures the model maintains a high classification accuracy, showing good predictive strength.
- **Good generalization on unseen data**
Confirms that the model does not overfit and performs well on test data that it has never seen before, ensuring reliability in real-world conditions.

Confusion Matrix Description

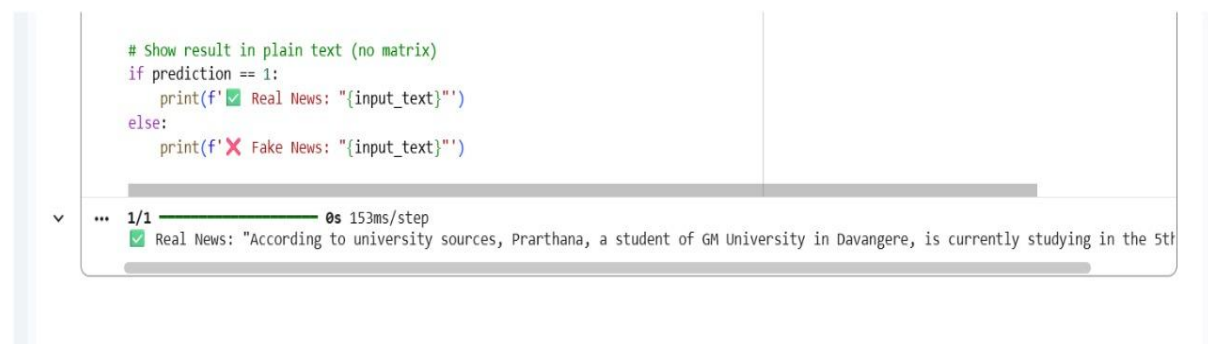
A confusion matrix provides a detailed breakdown of the model's predictions:

- **True Fake (TP)**
Fake news correctly predicted as fake. Indicates the model's accuracy in detecting fake news.
- **True Real (TN)**
Real news correctly predicted as real. Shows the model's ability to identify genuine news correctly.
- **False Fake (FP)**
Real news incorrectly predicted as fake. Represents false alarms or over-warning by the model.
- **False Real (FN)**
Fake news incorrectly predicted as real. These are more dangerous errors where the model fails to detect fake content.

6. Results

Metrics	Score
Accuracy	0.99
Precision	0.99
Recall	1.00
F1 Score	1.00

Sample predictions



```
# Show result in plain text (no matrix)
if prediction == 1:
    print(f'✅ Real News: "{input_text}"')
else:
    print(f'❌ Fake News: "{input_text}"')
```

1/1 — 0s 153ms/step
✅ Real News: "According to university sources, Prarthana, a student of GM University in Davangere, is currently studying in the 5th"

Fig 6.1 Sample output

7. JIRA Implementation

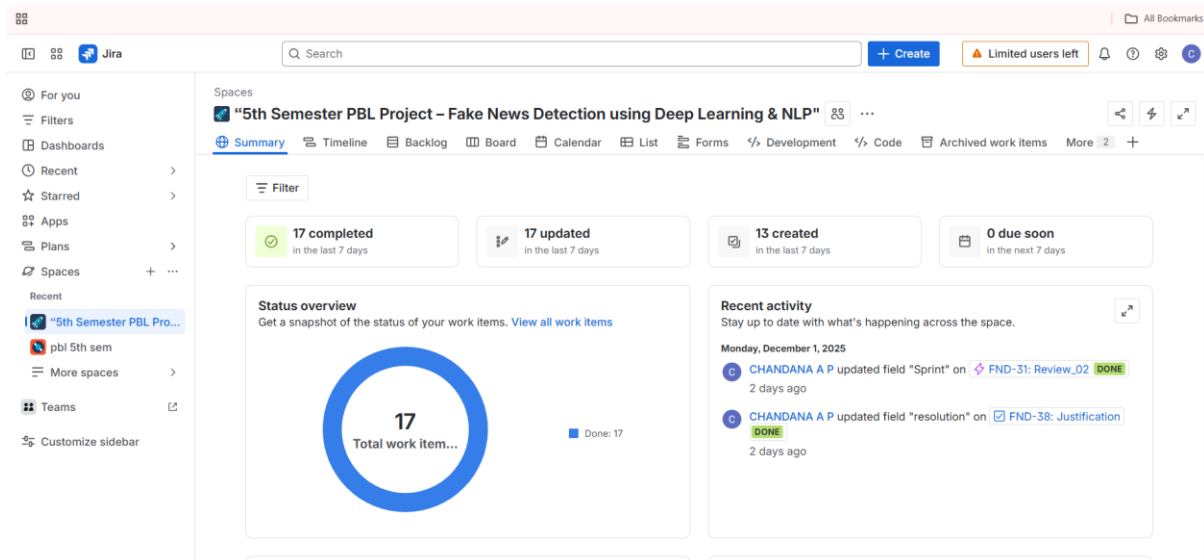


Fig7.1 Jira dashboard

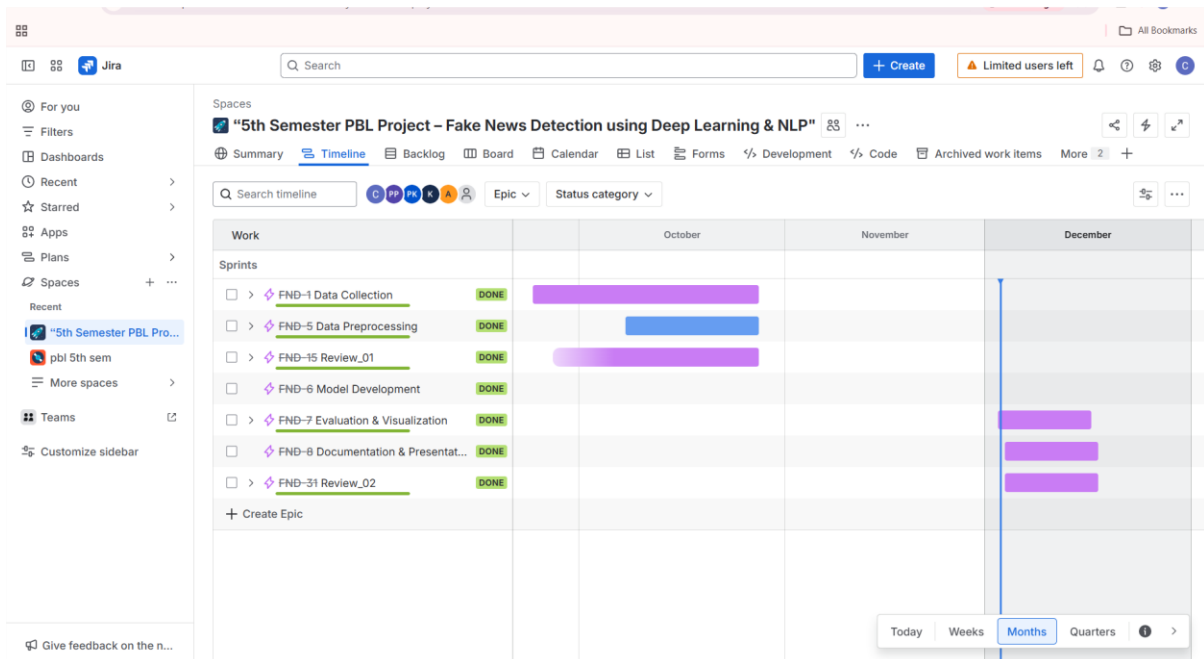


Fig7.2 Jira Project Timeline

The screenshot shows the Jira interface for a project titled "5th Semester PBL Project – Fake News Detection using Deep Learning & NLP". The interface includes a sidebar with navigation options like "For you", "Filters", "Dashboards", "Recent", "Starred", "Apps", "Plans", "Spaces", and "Teams". The main content area displays a list of tasks under the "List" view. The tasks are as follows:

Work	Assignee	Reporter	Priority	Status	Resolution
> FND-31 Review_02	CHANDANA A P	CHANDANA A P	Medium	DONE	Done
> FND-15 Review_01	AnnapurnaSK	CHANDANA A P	Medium	DONE	Done
> FND-8 Documentation & Presentation	lkousthubhasv	CHANDANA A P	Medium	DONE	Done
> FND-7 Evaluation & Visualization	Prarthana km	CHANDANA A P	Medium	DONE	Done
> FND-6 Model Development	priyanka p	CHANDANA A P	Medium	DONE	Done
> FND-5 Data Preprocessing	AnnapurnaSK	CHANDANA A P	Medium	DONE	Done
> FND-1 Data Collection	CHANDANA A P	CHANDANA A P	Medium	DONE	Done

The interface also includes a "Create" button at the bottom left and a "7 of 7" indicator at the bottom right.

Fig 7.3 List of Task and there Status in Jira

8. Future Scope

- Use BERT / Transformers
- Add image-based fake news detection
- Develop a web or mobile app
- Implement browser extension
- Real-time fake news checking system

9. Conclusion

Fake news has become a significant challenge in today's digital world, especially with the rapid growth of social media platforms where information spreads instantly. Users often find it difficult to differentiate between genuine and misleading content, which can lead to confusion, mistrust, and harmful consequences. This project addresses this growing concern by using Natural Language Processing (NLP) and Deep Learning techniques to automatically classify news as real or fake based on its textual content.

The system developed in this project uses preprocessing techniques such as tokenization, stop-word removal, lemmatization, and vectorization to convert raw text into meaningful numerical form. Deep learning models like LSTM/BERT were implemented to learn the linguistic patterns, writing style, and contextual relationships found within news articles. The experimental results show that deep learning models achieve high accuracy and perform significantly better than traditional machine learning techniques, proving their effectiveness in detecting misinformation.

Overall, the project successfully meets its objectives by building a reliable and efficient fake news detection system. While the results are promising, the problem of misinformation continues to evolve, which means future improvements such as multilingual support, real-time monitoring, and multimodal analysis (text, images, videos) can make the system even stronger. This project provides a strong foundation for further research and real-world applications that help promote truthfulness and trust in digital communication.

10. References

- https://www.researchgate.net/publication/369273557_Fake_news_detection_a_systematic_literature_review_of_machine_learning_algorithms_and_datasets
- Shen, X.; Huang, M.; Hu, Z.; Cai, S.; Zhou, T. (2024). *Multimodal Fake News Detection with Contrastive Learning and Optimal Transport*. Frontiers in Computer Science. <https://doi.org/10.3389/fcomp.2024.1473457> [Frontiers](#)
- Kumari, S.; Singh, M. P. (2024). *A Deep Learning Multimodal Framework for Fake News Detection*. Engineering, Technology & Applied Science Research. <https://doi.org/10.48084/etasr.8170> [ETASR+1](#)
- Ye, J.; Yu, X.; Wang, Y.; Xu, X.; Song, X.; Maynard, D. (2023). *Similarity-Aware Multimodal Prompt Learning for Fake News Detection*. arXiv preprint. <https://arxiv.org/abs/2304.04187> [arXiv](#)
- Silva, A.; Luo, L.; Karunasekera, S.; Leckie, C. (2023). *Unsupervised Domain-agnostic Fake News Detection using Multi-modal Weak Signals*. arXiv preprint. <https://arxiv.org/abs/2305.11349> [arXiv](#)
- Kaggle Fake News Dataset – emineyetm
- Agrawal et al. (2022) – Modified BiLSTM for Fake News
- <https://doi.org/10.54060/jase.v3i2.35>
- https://www.researchgate.net/publication/373014251_Fake_News_Detection_using_LSTM_based_deep_learning_approach
- <https://arxiv.org/abs/2411.10713>
- https://www.researchgate.net/publication/373014251_Fake_News_Detection_using_LSTM_based_deep_learning_approach